

Small Edges, Strict Boundaries: A Chronological Evaluation of Intraday Stock Direction Classification

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Abstract

In intraday five-minute stock-direction classification, a weak edge can survive strict evaluation yet hold only under conditions. We establish this with an evaluation protocol fixed before any model is scored. It freezes the label, no-trade band, chronological splits, roster, same-row dummy floor, and a counted validation-look budget. The controls matter because validation contact, leakage, and repeated comparison can inflate macro-F1 on a target that sits just above a stratified-dummy floor. On the frozen validation split, a predeclared temporal convolutional network exceeds the same-row stratified dummy by 1.69 pp in macro-F1 (two-seed mean, 1.63 pp at the worse seed; 2 seeds cannot quantify variance). On a separate guarded, non-independent walk-forward over post-2017 (expanding windows share rows), which is not a clean test, the same candidate stays positive at +0.636 pp row-pooled and clears a deliberately weak bar. A descriptive CSCV/PBO of 0.514 leaves overfitting unresolved. The candidate's edge is also conditional: it concentrates in the calmest bars and falls below the balanced random prior on the most active ones. This low-to-high pattern recurs across both windows, though we cannot yet separate it from a bid-ask-bounce artifact. We measure macro-F1, not economic value, on five large-cap US survivors with no known-signal positive control. The contribution is the evaluation discipline and a map of where a thin edge holds and where it inverts. Whether the protocol also shows such signs are absent on a genuine signal is the next test.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Mathematics of computing** → *Probability and statistics*; • **Applied computing** → *Economics*.

Keywords

intraday direction classification, chronological validation, no-trade band, model calibration, selective classification, backtest overfitting

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1 Introduction

A small directional signal in intraday five-minute stock prices can survive careful scrutiny, or it can be an artifact of how the evaluation was run. On a frozen validation split, a predeclared temporal convolutional network exceeds a same-row stratified-dummy floor by 1.69 pp in macro-F1, a margin that sits just above a near-even floor. Validation contact, temporal leakage, and repeated model comparison each manufacture gaps of that size [3, 17, 22], so the number alone settles nothing. We ask what remains of such an edge once the evaluation is made strict, and where it lives.

We answer with an evaluation protocol that fixes the terms of the test before any model is scored. It freezes the label, no-trade band, chronological day-block splits, model roster, same-row dummy floor, and a counted validation-look budget, then scores each model against that floor. The protocol makes no architecture claim and selects no final model; the model families exist to exercise it. Financial machine-learning studies fit flexible models to stock and limit-order-book prediction [15, 39], while a parallel literature shows how adaptive validation and backtest overfitting inflate the gains they appear to deliver [3, 17, 22]. We take the step before a trading backtest and ask what a return-free, drop-neutral direction-classification estimand yields under these controls.

Run under the protocol, that estimand yields a small and uneven edge. On the frozen validation split the predeclared TCN primary's two-seed mean margin over the same-row stratified dummy is 1.69 pp in macro-F1, dropping to 1.63 pp at the worse seed. 2 seeds are too few to quantify variance. The pooled margin's descriptive block-bootstrap interval excludes zero, and the gain holds as a point estimate in all five tickers but spreads widely. Train-inner control comparisons across four control rows (two of them last-step ablation controls) show a 0.66 pp spread, smaller than the 1.69 pp official-validation margin though not a full-family comparison. That spread sets the thin scale of the edge against a near-even floor.

Past the validation boundary, a guarded walk-forward applies the same discipline. This is not a clean test: the post-2017 segment is a non-independent confirmation of the validation-frozen primary, scored more than once within the project [8]. The predeclared primary clears a deliberately weak stability bar there, with a row-pooled delta of +0.636 pp. A descriptive multiplicity check returns a probability of backtest overfitting of 0.514, near a coin flip. The guarded readout corroborates the sign on its own segment; it does not strengthen the validation claim, and the three domains stay separate.

Because neither readout turns a thin margin into a strong one, the protocol's most informative outputs are two diagnostics, not the headline number: whether the confidence scores are usable, and where the edge concentrates. Calibration error is small while its resolution is near-zero (equal-mass ten-bin ECE ≈ 0.010), so the confidence scores cannot define an operating rule. Resolved by activity, the edge is largest on the calmest bars (low-activity tercile,

+5.43 pp over dummy) and falls below the balanced random prior on the most active ones (high-activity tercile, macro-F1 ≈ 0.483). The same low-to-high pattern recurs in the guarded 2017–2024 window. Here activity means eligible-row count under the no-trade band, not volume or liquidity. A bid-ask-bounce alignment we cannot rule out [29] keeps the pattern a diagnostic boundary rather than model-attributable predictability.

The paper therefore contributes an evaluation protocol, a near-null worked case that exercises it end to end, and a map of where a thin edge holds and where it inverts. The case is deliberately narrow: five large-cap US survivors, 2 seeds, and macro-F1 only, with no profit-and-loss, cost model, or known-signal positive control. What the protocol records here is the contribution: estimand sensitivity, a weak stability bar, multiplicity fragility, and a conditional sign that a headline-only readout would hide. Whether it also shows these signs are absent on a genuine signal is the positive-control next step.

2 Related Work

Machine-learning models for equity and limit-order-book prediction set the task context [11, 15, 39]. We borrow the task but narrow the estimand to drop-neutral, 5-minute direction classification scored against a same-row stratified dummy, so a macro-F1 margin here is classification evidence rather than investment performance.

The central methodological risk is contact between a model and its validation data. Work on leakage in machine-learning science and on decision-time backtest leakage shows how adaptive validation inflates apparent performance [22, 38]. Reusable-holdout and preregistration methods supply validity-preserving guards [8, 20], and chronological evaluation supplies the temporal one [21]. Finance controls data-snooping with backtest-overfitting probabilities, reality-check tests, and multiple-testing corrections [2, 3, 17, 26, 30, 33]. Our protocol assembles these guards for one return-free macro-F1 estimand: a fixed label and band, chronological day blocks with day-local windows, a same-row stratified dummy, and a counted validation-budget ledger. We exercise the assembly on a single near-null case rather than a known-signal control. Return-based benchmarks and investing-strategy studies score the later economic stage [24, 38]; our readout audits the classification gate upstream of it.

The model families are deliberately ordinary. Temporal convolutional networks, decomposition-linear models, and gradient-boosted trees are established sequence and tabular baselines [1, 14, 23, 37], and we use them to drive the protocol without an architecture claim or a final-model choice. Because the families are off-the-shelf, any margin that survives is evidence about the estimand and the protocol, not about a model design.

With the model axis set aside, two boundaries remain on what a margin can mean. The first is whether confidence scores rank cases usefully, where selective classification and calibration enter only as an adverse diagnostic [7, 12, 16, 32]. Brier-score decomposition separates calibration error from resolution [6, 27], so small calibration error need not imply useful ranking, and the ranking must be checked directly. The second boundary is microstructural. Prior intraday-return work documents cross-sectional periodicity, time-varying short-horizon predictability, and seasonality [4, 18, 19]. It

is a foil: that predictability tracks timing, active high-volume states, or seasonal rebalancing, not the calm slice where our edge concentrates. Bid-ask bounce, in turn, can create row-level structure that a within-day permutation leaves intact [25, 29]. Our controls cannot yet rule it out, so we keep the activity-tercile result as a conditional limitation.

3 Task and Evaluation Protocol

The protocol fixes the label, the splits, the baselines, the readout metric, and the decision rules before the first model is scored. Every margin we later report is therefore prespecified, never a post hoc selection. Split-boundary controls decide which rows are eligible to score, and a validation-budget ledger records each logged contact with the validation split behind that margin [22, 26]. One limit is built in: we exercise the protocol on a single near-null case with no known-signal positive control, so its power to separate signal from noise rests on construction here, not on a demonstration.

3.1 Task

The task is drop-neutral binary intraday direction classification on 5-minute bars, resampled from 1-minute data for five US large-cap equities (CSCO, JPM, KO, MSFT, WMT). One prediction unit is a single ticker at a single completed 5-minute target bar, scored up or down over a fixed forward horizon. Any edge the protocol preserves is classification evidence, not economic value.

3.2 Label

Each row's label is the sign of the 9-bar-ahead forward return outside a no-trade band. Writing θ for the band in return units, a row is labeled up when the forward return exceeds $+\theta$ and down when it falls below $-\theta$. Any row inside the band is invalid and is dropped. We set θ to 3.0 bps. The band removes near-flat moves whose direction is set by noise [31, 35] and shifts the label distribution away from the all-bars distribution [36]. It also defines the per-day activity proxy that a later conditional diagnostic uses, which is a limitation rather than a result. Labels and windows are day-local and per-ticker: no window and no label crosses a trading-day boundary or a ticker. The surviving up and down rows are the only supervised rows, and no-trade rows are neither a third class nor imputed.

3.3 Chronological splits

Chronological day blocks define the partitions and are fixed by the Stage 00 freeze [21]. Train runs from 1998-01-02 to 2013-09-16 (end exclusive), official validation from 2013-09-16 to 2017-01-25 (end exclusive), and the holdout is closed from 2017-01-25 onward. No stage in the validation pipeline reads or scores a post-2017 bar, and the run manifest records holdout contact as false. The guarded walk-forward in Section 7 opens that segment only after the model is frozen, so it confirms the frozen primary rather than testing it cleanly. Because every window and horizon stays inside one trading day, the last train target window and the first validation target window cannot straddle the cut or share a row. The template substitutes this within-day locality for purge and embargo, and we apply neither; cross-day serial dependence therefore stays intact, which is a limitation, not a solved problem. Preprocessing that

learns statistics is fit on train rows alone and applied unchanged to validation. The feature window $w = 20$ is likewise set by a train-inner feature-and-window search (Section 5), not tuned on the validation readout.

3.4 Baselines

Every model score is read against same-row trivial baselines on the identical target rows, within each evidence domain. The validation criteria use a stratified dummy that samples from the empirical train label prior and a majority baseline that always predicts the most frequent train label. The stratified dummy pins a same-row, label-prior macro-F1 floor for the drop-neutral target. Reading every model on the identical rows against this prior closes the class-imbalance illusion: a margin over it means “above train-prior guessing,” not a score inflated by base rates. That margin still does not by itself exclude serial structure or microstructure effects, which later guards and diagnostics address.

3.5 Readout metric and predeclared criteria

We score the model-versus-dummy margin in macro-F1 on the drop-neutral up/down target. A dated protocol fixed the decision criteria before any readout [20], and they apply to the predeclared TCN primary. On the validation split the primary passes when its macro-F1 delta is positive over both the stratified dummy and the majority baseline. At least three of the five tickers must also be positive, where three is a floor, not the achieved count. These floors are deliberately weak: predeclaration fixes the bar in advance, but it does not make the bar discriminating. The guarded walk-forward carries its own separate and also weak predeclared stability bar. Neither readout selects a model, and no final model is selected.

3.6 Validation-budget ledger

A validation-budget ledger counts and tier-tags every readout contact with the official validation split, and each contact carries for_selection set to false. The ledger makes logged route contacts traceable and bounds disclosed reuse of the validation split; it is an internal count, not an external timestamp. It later records 2 official-validation seed-candidate scoring events against 56 guarded scoring events. The protocol keeps three evidence domains separate and reports each number only within its own domain. Those domains are the official validation readout over 2 seeds, the train-inner control comparisons, and the guarded non-independent walk-forward readout, also scored under the two prespecified seeds. Every reported number names its domain, and no sentence mixes them.

Figure 1 summarizes the timeline and the freeze ordering.

4 Models

The roster exists to drive the protocol, and only the TCN is primary. We score four families: a temporal convolutional network, a decomposition-linear model, a decomposition-convolutional hybrid, and gradient-boosted trees [1, 14, 23, 37]. None is a known-signal positive control. The roster therefore exercises the protocol on a near-null case: it cannot prove that a surviving edge is not manufactured by an uncontrolled confound, nor that the protocol would catch a real signal.

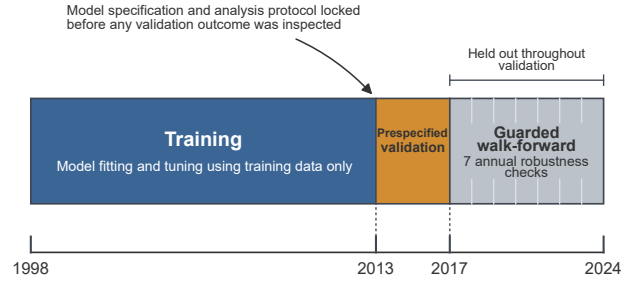


Figure 1: Protocol timeline and freeze ordering for the drop-neutral up/down direction task. Chronological day-blocks define training (1998-01-02–2013-09-16), one official validation readout (2013-09-16–2017-01-25), and a guarded post-2017 era. Model specification and analysis protocol are locked before validation; validation is prespecified, not tuning or model selection, and no final model is chosen. The post-2017 region was untouched during validation; the guarded walk-forward checks confirm a frozen model over non-independent periods, so they are not future-blind or independent holdout tests.

The predeclared primary is a temporal convolutional network [1] in a `tcn_tiny` configuration: two residual blocks of 16 channels each, kernel size 2, and learning rate 10^{-3} . Standard DLinear (Std-DLinear) splits each input window into trend and residual components and feeds each to its own linear head before summing them [37]. The hybrid pairs a multi-scale DLinear branch (MS-DLinear) with a small residual TCN branch (together, MS-DLinear+TCN) and is a roster alternative, not a contributed model. Its MS-DLinear branch combines several decomposition kernels in the manner of series-decomposition forecasters [34, 40]. The fourth family is LightGBM [23], a gradient-boosted tree run as a tabular baseline on windowed features [14].

Predeclaration covers the roster and the primary-versus-fallback rule, and the train-inner search completes before either readout opens. The search fits the TCN configuration above and holds LightGBM in reserve as the primary-slot fallback for mechanical or reconstruction failures. None occurred, so the fallback never fired. The frozen specification then enters both readouts unchanged, and neither readout selects a final model. On the guarded readout some roster alternatives post numerically higher deltas than the TCN primary, but only post hoc and not like-for-like; the TCN is the lone pre-committed frozen primary. No recurrent model enters the roster, and the last-step multilayer perceptron (`last_step_mlp`) appears only as a train-inner control, never as a scored family. The four families share the data, seeds, and frozen designation that the next section fixes.

5 Experimental Setup

This section fixes the data and the train-inner search deferred from the protocol section, including the window and feature set left to that search; Table 1 lists the frozen statistics. The study uses five large-cap US equities (CSCO, JPM, KO, MSFT, WMT) as 5-minute bars resampled from 1-minute data. The chronological split places 736,685 labeled target rows in the train interval (1998-01-02 through

Table 1: Frozen dataset statistics. Counts are eligible binary target rows after the no-trade band invalidates near-flat rows, over end-exclusive chronological day blocks. The hold-out from 2017-01-25 received zero validation-phase contact (holdout_test_contact=false).

Property	Value
Tickers	CSCO, JPM, KO, MSFT, WMT (5)
Frequency	5-min bars (from 1-min)
Train interval	1998-01-02 – 2013-09-16 (excl.)
Validation interval	2013-09-16 – 2017-01-25 (excl.)
Train rows	736,685
Validation rows	151,064 ($\approx 29.9\text{k} - 30.8\text{k}/\text{ticker}$)
Label horizon	9 bars
No-trade band	3.0 bps
Window w	20
Primary feature set	price_volume_time
Fallback feature set	price_action_core

2013-09-16, end exclusive) and 151,064 in the official validation interval (2013-09-16 through 2017-01-25, end exclusive). Per-ticker validation counts span $\approx 29.9\text{k} - 30.8\text{k}$. Labels use a 9-bar horizon and a 3.0 bps no-trade band. The input is a 20-bar window over the price_volume_time feature set, the output of the train-inner feature-and-window search; price_action_core is the registered fallback feature set. This step shortlists only the feature set and window; family designation is the separate train-inner step below.

The separate train-inner step fixes the predeclared primary-and-fallback designation and one profile per family. The frozen feature-window and train-inner configs, the handoff, and the readout records bind the grid, the search-space files, and the inner folds. Those same records also fix the selected profiles, the refit rows, the early-stopping source, and the resolved device. This bounded-profiles search is a screening-grade train-inner selection, not an unconstrained or validation-tuned hyperparameter search. The official validation readout only characterizes the frozen specification and is not a model-selection event; the route manifest sets no_final_model_selected=true. We then report the official validation outcome in macro-F1 over $n=2$ seeds (101 and 202), refit on the recorded CUDA device. Two seeds are too few to quantify seed-to-seed variance.

6 Validation Results

We now read the predeclared TCN primary on the frozen validation split. Every number in this section comes from that split over $n = 2$ seeds, unless a sentence names a different evidence domain. Two seeds yield a descriptive spread rather than a variance estimate. The readout characterizes the frozen candidate and ranks nothing; the route declares no final model.

The primary meets the predeclared criteria. The predeclared TCN primary reaches macro-F1 of 0.5170 ± 0.0009 across 2 seeds, where the $\pm$ value is the standard deviation alone (Table 2). It exceeds the same-row stratified dummy by 1.69 pp in the seed mean, with the smallest per-seed margin at 1.63 pp, and it exceeds the majority-class baseline by 18.8 pp. The per-ticker delta is a positive cross-seed point estimate for all five tickers. The three predeclared conditions

are met. That is a fact about a fixed bar, not a demonstration of effectiveness: the margin is a thin validation-domain edge above a near-even floor.

The floor sets the scale. The dummy baselines define what “thin” means. The same-row stratified floor is roughly 0.499–0.501 in macro-F1, close to the even-class value, while the majority-class baseline collapses to 0.329 by abandoning one class entirely. The 18.8 pp gap over majority therefore reflects the stratified dummy’s two-sided coverage rather than a strong directional signal; the weak-signal reading rests on the 1.69 pp gap over the stratified dummy.

A train-inner control spread, not a family comparison. A second evidence domain frames the size of that margin. Train-inner control comparisons across four control rows, two of them last-step degenerate controls, show a 0.66 pp spread in macro-F1, smaller than the 1.69 pp official-validation margin of the TCN primary over the same-row stratified dummy. The four rows are train-inner diagnostic controls, not the guarded roster’s full model families, so the 1.69 pp margin and the 0.66 pp spread sit in separate domains and form no apples-to-apples pair. The comparison only sets scale; it supports the evaluation-discipline framing and ranks no model family.

The primary is a TCN, and the fallback never fires. The predeclared primary is the TCN configuration on the price_volume_time feature set with window $w = 20$. The LightGBM fallback stays inactive: it is reserved for mechanical or reconstruction failures, none of which occurred, so no fallback path enters the readout.

The per-ticker edge is positive but uneven. The per-ticker deltas are all positive as cross-seed point estimates, yet they spread widely (Figure 2, Panel A). CSCO carries the largest cross-seed mean delta at +2.21 pp and JPM the smallest at +1.00 pp, and the JPM block-bootstrap interval crosses zero. The pooled 1.69 pp margin is thus an average over a heterogeneous panel, which is one reason the diagnostics in Section 8 ask where the same-row edge concentrates and where it inverts below the random prior.

How the bands and pooled margins are computed. The capped whiskers in Figure 2 are unadjusted 95% per-trading-day block-bootstrap intervals conditional on fixed seeds. Whole-day resampling handles within-day autocorrelation but not between-day regime clustering or cross-ticker common-factor dependence. The per-group target-row counts (roughly 30k in Panel A validation tickers and up to 55k in Panel B guarded periods) are therefore not independent samples, and the bands reflect scored-row sampling only. The first row in each panel is row-pooled, not averaged: per seed, pool target rows within the domain, recompute candidate and stratified-dummy macro-F1, take the delta, then average the two seed deltas.

7 Guarded Walk-Forward Readout

What the guarded readout is. Past the validation boundary, we run a guarded walk-forward readout over seven consecutive 12-month periods on the post-2017 segment. The readout confirms the already-frozen primary over expanding-window periods that share rows and a fixed survivor universe, with no final model selected, so

Table 2: Frozen validation split, predeclared TCN primary: seed means over $n = 2$ seeds (mean $\pm$ std where shown). Deltas are pp macro-F1 changes against named same-row baseline.

Quantity	Value
TCN primary macro-F1	0.5170 ± 0.0009
Δ vs. stratified dummy (seed mean)	+1.69 pp
Δ vs. stratified dummy (min seed)	+1.63 pp
Δ vs. majority class	+18.8 pp
Stratified dummy macro-F1	$\approx 0.499-0.501$
Majority-class macro-F1	0.329
Per-ticker Δ vs. stratified dummy (cross-seed mean)	
CSCO (max)	+2.21 pp
JPM (min)	+1.00 pp
Positive-ticker count	5/5

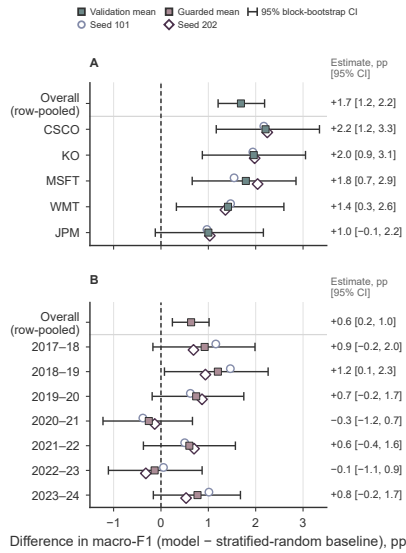


Figure 2: Descriptive macro-F1 deltas for the predeclared TCN primary against same-row stratified dummy; positive values favor the model. Panel A is prespecified validation by ticker; Panel B is the guarded 12-month walk-forward. The JPM interval and six of seven guarded intervals include zero. Panel B is not a clean test, independent holdout, or model-selection ranking.

it is a non-independent confirmation rather than a future-blind test. The post-2017 segment was also scored more than once within the project. An earlier walk-forward run preceded the bound re-run reported here, so the holdout is spent and its second reading is a reuse, not a fresh look [8]. For both reasons the readout makes no clean-test, out-of-sample-proof, or final-model claim. The other roster rows are comparison evidence, never ranked and never selected.

The bar is a weak floor, not the observed count. The predeclared TCN primary met the predeclared guarded stability criteria. The criteria are an intentionally weak floor: at least two of seven periods positive in macro-F1 over the same-row stratified dummy, and a positive pooled delta. Under an independent fair-coin null the

two-of-seven clause alone passes about 94% of the time, so the bar only prevents post hoc relaxation. The observed five-of-seven positive periods (Figure 2, Panel B) are descriptive context, neither certification nor the bar. Because the edge is near-null, we read the guarded readout as one worked instance of the protocol, not as evidence of a durable edge.

Behind that pass the period-level deltas are uneven. The readout comprises 56 guarded scoring events (seven periods, four families, 2 seeds), judged on the primary's rows alone; the addenda only re-aggregate the frozen per-period prediction dumps from the predeclared walk-forward refits. The primary's 2020-21 and 2022-23 period means fall in the COVID and bear-market windows, and both are negative. The 2022-23 period also reverses sign across the two seeds. With only 2 seeds we cannot quantify that dispersion, so the per-period readout is not even seed-stable there.

The headline delta is estimand-sensitive. The headline delta moves with the aggregation choice, so we report both units. The binding quantity is the row-pooled delta over the protocol row-union, +0.636 pp in macro-F1 over the stratified dummy; the equal-weight-per-period companion is +0.550 pp. Because the bar tests only the sign of a positive pooled delta, it is insensitive to this choice. The headline magnitude stays small under either unit.

Repeated comparison across the roster could also inflate this small edge, so the protocol applies a multiplicity discount on the binding row-pooled estimand. We use an odd-block CSCV/PBO adaptation ($n_{\text{trials}}=4$, $n_{\text{blocks}}=7$), not the canonical symmetric form; these are descriptive discounts, not significance tests [2, 3, 17].

Pre-registration, not peak performance, survives multiplicity. Across the four-family by seven-period guarded roster the CSCV probability of backtest overfitting is 0.514, near a coin flip. The numerically largest family mean belongs to LightGBM at +0.726 pp, yet its period lower bound is negative (-0.030 pp). Only the predeclared TCN primary has a period lower bound above zero (+0.047 pp). Under the multiplicity discount the predeclared candidate, not the highest scorer, is the one that survives, and no family is selected. Consistent with that, the validation-budget ledger keeps the contact asymmetry explicit: 2 official-validation scoring events against 56 guarded events, all with `for_selection=false`.

The guarded sign is small but stable. On its own segment the guarded sign is small but stable. Its pooled block-bootstrap interval, [0.24, 1.02] pp, excludes zero as a descriptive resampling interval rather than a significance test. The sign survives every leave-one-period-out and leave-one-ticker-out drop (worst cases +0.538 pp and +0.514 pp). A within-day label-shuffle sentinel collapses the delta toward a negative null, so the edge is not a within-day-leakage artifact. These checks are descriptive and stay under the multiplicity discount above. What survives strict evaluation here is therefore a predeclared sign, not a strong number: the direction holds weakly on its own segment, and the predeclared candidate clears multiplicity where the highest scorer does not. A stable pooled sign can still hide a direction that flips across conditions, and Section 8 resolves the same-row edge by activity to find that boundary.

8 Diagnostics and Robustness

The diagnostics put two limits on the readouts. The confidence-based acceptance signals of a near-base-rate classifier are largely illusory, and the thin edge inverts across activity terciles.

Calibration error is small, but resolution is near zero. On the frozen validation split the pooled probability output has an equal-mass ten-bin expected calibration error of about 0.010. The Brier score is 0.2496, but its vector decomposition is dominated by the uncertainty term (0.2499), leaving a resolution component near 4.7×10^{-4} [6, 27]. Small calibration error therefore does not mean the confidence scores rank cases usefully [16], so the ranking has to be tested on its own terms.

Selective prediction barely improves on random ranking. On the same split the accuracy-based risk-coverage curve (Figure 3) carries no cost model and marks no operating point. Its area (AURC) is about 0.470 against an oracle near 0.140, an excess area (e -AURC) of about 0.330 in the seed mean. These are within-seed, day-level summaries rather than seed-level estimates over 2 seeds, and the gain reads thin. Against the random-ranking reference the predeclared score lowers AURC only marginally (Δ AURC = -0.010 ; 95% trading-day cluster-bootstrap CI [$-0.014, -0.006$]). That is about 3% of the gap between random ranking and a perfect-ranking oracle (gap-closed 95% CI [$1.9\%, 4.0\%$]). The gain shrinks further at high coverage. The mean random-minus-score selective-risk difference over the 80–100%, 90–100%, and 95–100% bands is only 0.25, 0.14, and 0.07 percentage points. The companion AUGRC is 0.237 [9, 12, 13, 32]. Because low-confidence rows tend to fall in the high-activity tercile, abstention removes the very rows where the below-random high-activity limitation is most visible. Abstention therefore compounds that limitation rather than fixing it [7].

Component ablation moves little (train-inner domain). Before slicing the edge by condition, we check in a separate domain whether it hinges on which component we credit. One train-inner control precedes the conditional split. Here `tcn_only` is the primary's TCN and `dlinear_only` is the hybrid's multi-scale DLinear branch (MS-DLinear) with its residual TCN removed. Each control row's gap to the reference stays within ± 0.7 pp of macro-F1 (`tcn_only` +0.03 pp, `dlinear_only` -0.14 pp, `last_step_lightgbm_control` -0.12 pp, `last_step_mlp` -0.63 pp; the last two are last-step degenerate controls). We keep this control inside its own domain, apart from the validation and guarded readouts. The edge is insensitive to the control-row choice, so it is not an artifact of which component we credit, which leaves open where in the data it actually lives.

The edge concentrates in calm bars. To find where it lives, we slice the same-row macro-F1 delta by activity terciles, the per-(ticker, trading-day) eligible-row-count proxy under the 3.0 bps no-trade band, not volume, liquidity, or volatility. The terciles are rank-defined within ticker, so there is no single global cut point. On the validation split the delta runs +5.43 pp, +1.91 pp, and -1.54 pp from low to high activity tercile (Table 3). In the high tercile the macro-F1 (≈ 0.483) sits below the even-class prior (≈ 0.498) and the same-row stratified-dummy floor (≈ 0.500). The same low-to-high sign pattern recurs in the guarded 2017–2024 era at +4.08 pp,

Table 3: Activity-tercile condition map. Validation uses 2 seeds and guarded is non-independent. The high tercile falls below the balanced random prior in both domains; block-bootstrap checks are descriptive. Seed-mean rows (low/medium/high): 44,144/51,572/55,348 validation and 95,310/111,178/119,479 guarded.

Activity tercile	Validation Δ	Guarded Δ
Low activity	+5.43 pp	+4.08 pp
Medium activity	+1.91 pp	+0.56 pp
High activity	-1.54 pp	-2.10 pp
High-tercile macro-F1	≈ 0.483	0.480
Balanced prior	≈ 0.498 – 0.500	

+0.56 pp, and -2.10 pp, with a high-tercile macro-F1 of 0.480 (Figure 4). The figure's bars are across-seed means with open circle and diamond seed markers. Its grey sleeves are per-trading-day block minimum-detectable effects from official validation or measure-only re-aggregation of frozen guarded dumps, not standard errors or seed-level uncertainty. This is the only axis we could re-measure on the guarded segment, and it is a non-independent re-aggregation, not an independent replication. The magnitude is small, but the low-to-high pattern is same-signed in both windows. The signal is thus absent where the classifier faces its hardest slice, the most active bars.

Base-rate and within-day artifacts are excluded; a microstructure one is not. A stable but small pattern invites the question of whether it is spurious, and three artifact families bear on it: two are ruled out and one is not. The near-constant per-tercile dummy floor (≈ 0.500) excludes a class-imbalance artifact, while the up-rate shows only a small monotone gradient (0.523, 0.515, 0.502 from low to high, about 2.1 pp). A fifty-permutation within-day label-shuffle sentinel excludes within-day leakage. The observed delta exceeds the per-tercile shuffle maximum in every tercile, including the high tercile, where it exceeds a more negative shuffle null while staying below random. Even the below-random value is therefore not a permutation artifact. The remaining alternative is microstructure. The calm-bar concentration cannot presently be separated from a bid-ask-bounce alignment, which is a true row-level feature-label coupling that a within-day permutation preserves [25, 29], and the sentinel has no power against it. The discriminating control needs raw five-minute bars the dump does not carry. We therefore report the map as a diagnostic boundary on this estimand, not as model-attributable conditional predictability. If it were model-attributable, it would run against prior intraday-return work tying short-horizon predictability to intraday timing, active high-volume states, or seasonal rebalancing [4, 18, 19]. A recent regime study reports forecastability in low-financial-uncertainty periods [10], a regime-level parallel rather than one at our resolution. We carry the deferred microstructure control as a limitation (Section 9).

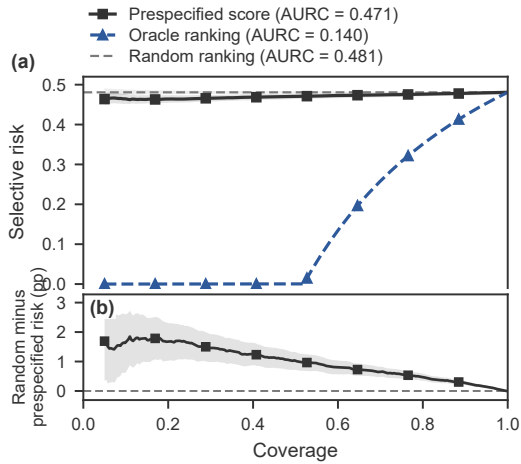


Figure 3: Accuracy-based selective-risk-coverage diagnostic; lower is better. The curve barely separates from random ranking, indicating weak confidence ordering rather than a usable abstention rule. There is no chosen operating point; with 2 training seeds, the plot establishes no statistical significance, calibration, or deployable abstention policy.

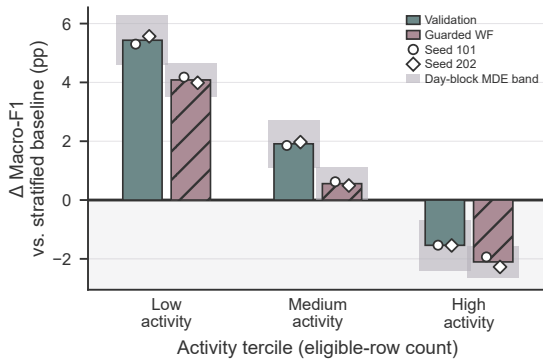


Figure 4: Activity-tercile condition map. Paired diverging bars compare validation and guarded macro-F1 deltas over the same-row stratified dummy; positive values favor the model. Activity is an eligible-row-count proxy under the no-trade band, not volume, liquidity, or volatility. The high-activity inversion is a limitation; guarded results are non-independent, not a clean test, and grey sleeves are not confidence intervals.

9 Conclusion and Limitations

The contribution is an evaluation protocol with three working parts. A pre-score freeze, same-row dummy floors, and a budget ledger keep the readout auditable whatever the outcome. Under that protocol the predeclared frozen TCN primary holds a small margin over its same-row baselines, consistent in pooled sign but uneven across tickers and regimes, which is an evaluation finding, not an architecture claim. Calibration and regime diagnostics then bound and interpret that margin, and the confidence scores rank cases too

weakly to define an operating rule. We name the magnitudes to be scrutinized rather than sold. The 1.69 pp validation margin and the 0.636 pp guarded row-pooled delta over the same-row dummy are thin edges above a near-even floor, and the backtest-overfitting probability is a near-coin 0.514. The validation margin even flips sign across activity, from 5.43 pp on the calmest bars to 1.54 pp below the dummy on the most active, a conditional sign a headline-only readout would omit.

The case is deliberately narrow. The protocol runs on one near-null case with no known-signal positive control, so its sensitivity to a genuine signal is untested. The official validation readout rests on 2 seeds, too few for seed-to-seed variance [5, 28], and the Student-*t* lower bound shipped with the decision record is provenance, not a decision criterion. The scope is a single market and five large-cap survivors of the 1998–2024 span, so any positive readout is entangled with survivor selection and does not extend to a broader panel.

The remaining limitations concern construct and claim scope. The no-trade band (3.0 bps) and the label horizon (nine bars) are pre-registered and frozen, with no threshold or horizon sensitivity scan. macro-F1 is a classification readout, not economic value, and with no execution model or transaction-cost accounting no margin here, validation, train-inner control, or guarded, is a trading claim. The guarded walk-forward is a non-independent confirmation, not a clean test, and no final model is selected. Its family margins are fragile. Under the descriptive multiplicity discount on the binding row-pooled estimand, the discounted per-period lower bound falls below zero for every family except the predeclared primary. That is not a like-for-like ranking, since the non-primary rows are post-hoc maxima and the primary a frozen profile, and the descriptive PBO is too coarse to rule overfitting in or out. The calm-bar concentration of Section 8 carries one open risk. The closing price close[*t*] enters both the most-recent-return feature and the forward-label denominator. A bid-ask bounce, largest relative to the small returns of the calmest bars, is therefore an alternative this classification-only evaluation cannot quantify [25, 29]. The label-shuffle sentinel rules out within-day leakage and day-base-rate artifacts but has no power against that bounce. The discriminating control, a half-spread proxy against the 3.0 bps band, needs raw five-minute bars the dump does not carry, so we defer it.

The defined next step is a known-signal positive control: confirming that the same protocol also shows these signs are absent when a real signal is present. Until then the outcome is a weak, conditional readout, not a model or a trading system.

Generative AI Usage Statement

Generative AI tools were used to assist with literature search, drafting and translating prose from the authors' notes, editing for style, generating figure-plotting code, and reviewing claim consistency, citations, provenance, and final-form issues. All experimental design, code execution, results, and claims were produced and verified by the authors. All AI-assisted text was reviewed, fact-checked against the project's experiment artifacts, and revised by the authors, who take full responsibility for the content.

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